Exercise 6

A walker that falls

At the end, the walker falls: it walks off a new ledge and lands on the floor.

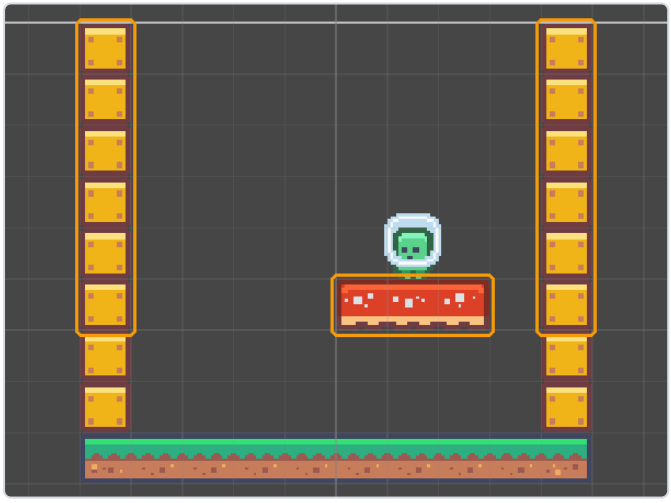
Before you start

- `PlatformerLab` with `MovementLab` from Exercise 5, where the walker walks and faces the way it goes; in a terminal, in the project folder, run `git pull` first
- `PlatformerPalette` from Exercise 3, with the red platform tiles `tile_0013` to `tile_0015`

1 A ledge and higher walls

Gravity needs something to fall from: a ledge to walk off, with a gap beside the right wall to fall along. Taller walls keep a jumping walker in the room.

- **Window** › **2D** › **Tile Palette**, check that Active Tilemap reads `Ground` and take the Brush (`B`).
- Paint a three-tile ledge two cells above the floor, over the sixth to eighth floor tiles (cells x = 0 to 2, y = 0): `tile_0014`, `tile_0013` and `tile_0015` are its left end, middle and right end. One empty cell stays between the ledge and the right wall.
- Pick the crate (`tile_0009`) and, with Box Fill (`U`), raise each wall by six crates, up to the cell at y = 5.
- Select `Main Camera` and set Position Y to `1`: the view now covers y = -4 to 6, so the walls reach its top edge and a jump from the ledge stays in view.
- Select `Walker` and set its Position to `(1.5, 1, 0)`: the ledge's top is at y = 1, and with the pivot at its feet the walker stands on it.



Six new crates on each wall, the ledge, and the walker standing on it.

2 Gravity

Gravity Scale 4 gives a quick, game-like fall. The velocity line from Exercise 5 keeps the vertical speed that physics gives it, so the walker can still steer in the air.

- On `Walker`, set Rigidbody 2D, Gravity Scale to `4`.
- **Play**, click the Game view and hold `A`: the walker walks off the ledge, starts falling in mid-stride and lands on the floor. Stop.
- If it falls through the floor, check `Ground`'s collision components from Exercise 4; if it doesn't fall at all, the velocity line sets y to a number instead of keeping `rb.linearVelocity.y`.
- `Ctrl` + `S`, then commit and push with the message `Add a ledge and gravity`.